A child lives with the young

The author's rule, what it settles, and the one place the code disagrees with it

13 September 2026 • stylized model

1 The rule

> *"maybe the right way to do this is to have the parent to have a child only when they are young and have no kids when they are old. then children are only with the young! and the young transmit the SUD penalty only if they get it in the transition to being old. The old dont weight the child at all since it left, so they only care about their own consumption and not the child's."*
>
> *"if the old die with SUD, that does not transmit to the child, since the child is already young at that time."*

Three statements, and they are one statement. A child is in the household for exactly one period, the parent's period 1. It leaves at the end of it, and it leaves as a young adult, which is why the cohort arriving each period has mass one. Everything that happens to the parent after that happens to somebody the child no longer lives with. So:

1. the equivalence scale is $\xi = \sqrt{2}$ in period 1 and 1 in period 2;
2. the old carry no child term — they value their own consumption alone;
3. the last moment that can transmit anything to the child is the *transition* out of period 1, where the disorder is realised. A parent who dies in period 2 with the disorder transmits nothing, because the child was already an adult when it happened.

2 The demography closes exactly

Each young household has one child. At a date there is mass 1 of children, mass 1 of young, and mass $\bar{s} = \phi_0(1 - \kappa\bar{\pi}) = 0.9067$ of old. This period's children are next period's young, so the cohort entering is mass 1 forever: no child is ever unhoused between households, and no period of a person's life is spent in nobody's household. That is the accounting the rule buys, and it is what fails if a child straddles the transition — then a parent's death truncates a childhood and the childhood flow has to be survival-weighted.

3 The weights add up, which is why the child's weight is $\sqrt{2}-1$

A household of two counts as $\xi=\sqrt{2}$ adult equivalents and reaches the utility of a single person holding $1/\xi$ of its resources. Give the parent weight 1 and the child weight $\xi-1$ and the household's weights sum to ξ exactly: with no drugs, $X_P=X_K=X$ and

$$u(X_P)\;+\;(\xi-1)\,u(X_K)\;=\;\xi\,u\!\left(\tfrac{c}{\xi},\tfrac{h}{\xi}\right),$$

so nothing is created or lost by the household having a child rather than not. A weight of 1 for the child would break that identity: it would count two people at full weight while deflating their bundle as though there were only $\sqrt{2}$ of them. $\xi-1=0.41421$ is the weight the scale already implies.

4 The childhood welfare formula that follows

The child's bundle in the parent's household, with the Stone–Geary shifter alone because the child consumes none of the drug:

$$X_K\;=\;\left(\tfrac{c}{\xi}\right)^{\varphi_c}\left(\tfrac{h}{\xi}\right)^{\varphi_h}\bar{b}^{\,\varphi_q},\qquad u(x)=\tfrac{x^{1-\sigma}}{1-\sigma}.$$

Childhood lasts one period and ends before any mortality is realised, so it carries no survival weight. A cohort's lifetime utility is

$$\boxed{W_t\;=\;(\xi-1)\big[u(X_{K,t-1})+\aleph\big]\;+\;\beta\,V_1(y_t)}$$

where $X_{K,t-1}$ is averaged over the period-1 distribution of parents at each household's *realised* allocation — a parent who buys drugs contributes its using allocation — and V_1 is the child's own adult value at the income its inherited penalty leaves it. The childhood is lived under the policy in force at $t-1$ and the adulthood under the policy at t; that asymmetry is the whole of the transition. Two notes on the boxed line. The $\aleph$ term is a switch: a childhood is a period of life, but $\aleph$ is added once per household in the Fortran rather than once per member, so counting it for the child too is arguable in both directions. And the β is currently absent from the code, which adds `adult + kid`; at this calibration $\beta=1.0$, so the omission has never shown up in a number.

5 Three of the four parts are already built

what the rule requires	where it lives	state
$\xi=\sqrt{2}$ in period 1, 1 in period 2	`stylized.py:882 _flow`	built
the old carry no child term	`stylized.py:972`, `child=False`	built
the SUD transmits on the transition out of period 1	`transition.py:119`, `use`×`prog`	built
nothing after the child leaves transmits	`transition.py:119`, `ev`	**contradicted**

The last row is the substance. The child's penalty has two legs, and they key on different periods:

- **the SUD leg** is `use_prob` × `prog`: the chance the parent uses in period 1 times the chance a user acquires the disorder at the transition. That is exactly the rule, already.
- **the homelessness leg** is `terminal_homeless_prob`, which is the parent's chance of being on the street *in period 2*. Its docstring says so in as many words: "THIS IS WHAT THE CHILD'S PENALTY KEYS ON ... It is when the household dies." Under the rule above, that is precisely the state that cannot reach the child.

This is a live contradiction with a standing instruction. Period-2 keying was not an accident: it was put there so that Treatment First's own revocations could reach a child, since the voucher is withdrawn at the transition and period-1 homelessness therefore cannot register it. Keying the leg on period 1 gives that up.

It does not, however, switch conditionality off. Treatment First still reaches the child through the SUD leg, because it changes how much people use and hence `use` × `prog`. What is lost is only the homelessness half of the revocation. **This is a modelling call, not a bug to be closed, so nothing has been changed.**

6 "SUD transmission is smaller" — where that bites

It does not bite in the stylized model. Drugs are chosen in period 1 only and the disorder is realised at the transition, so there is no old-acquired SUD to remove: the transmitted object is already the period-1 one. The share of children born behind the SUD penalty stays at 3.86 per cent.

It bites in the *quantitative* model, where drug use is chosen at every age and a parent can acquire the disorder long after its children have left. There the rule genuinely cuts the transmitting population, and by construction it cuts it to the parents who acquired the disorder while the children were still at home. One consequence to expect rather than to be surprised by: δ_d, the size of the penalty, is calibrated to match the child's SUD odds ratio, so a smaller transmitting share is met by a *larger* fitted penalty. The share falls; the product is pinned by the target.

7 The numbers this rests on

the child in the household		**what the child inherits**	
equivalence scale ξ	1.41421	participation	0.08871
child's weight $\xi - 1$	0.41421	of users, acquire the disorder	0.41348
parent's weight	1.00000	population SUD rate	0.03668
household total	1.41421	children with the SUD penalty	0.03863
		children with the street penalty	0.00860
the childhood being lived		income shift, SUD penalty	0.56042
mean consumption c	0.66747	income shift, street penalty	0.69000
mean housing h	0.26338	income shift, both	0.38669
share on the street	0.01341		
share whose parent uses	0.08695	**the levels**	
$u(X_K)$	-7.54356	childhood, goods $(\xi - 1)u$	-3.12465
$u(X_P)$	-7.53432	childhood, $(\xi - 1)\aleph$	4.55813
		childhood, total	1.43349
masses at a date		adult value V_1	8.19940
children	1.00000	lifetime	9.63289
young	1.00000	childhood's share	14.88%
old	0.90672	what the parent itself books, $\omega_c(\xi - 1)u(X_K)$	-0.04743

For scale: one per cent of consumption in every state is worth 0.0613 in these units at the adult value. The parent's own valuation of its child, $\omega_c(\xi - 1)u(X_K) = -0.047$, is therefore below one per cent of consumption — ω_c calibrated to 0.0152 is what makes the damage to the child close to a pure externality, which is the point of the parameter.

Parameters: $\varphi_c = 0.48189$, $\varphi_h = 0.37355$, $\varphi_q = \theta = 0.14456$, $\sigma = 2$, $\aleph = 11.00431$, $\bar{b} = 0.05785$, $\omega_c = 0.01518$, $\beta = 1.0$, $\phi_0 = 0.91$, $\kappa = 0.09839$, $\delta_d = 0.57906$. Calibrated vector `calibrated_sep13.json`, iteration 1350. *β came to rest at* 1.0*, which is its upper bound and which* `stylized.py` *records as having been tried and rejected on an earlier calibration; flagged, not acted on.*